

Improving Preparedness of Communities for Evacuations using ZEVs

Project Overview

Meeting #1 with Community Partners

Ricardo de Castro (UC Merced)

April 26th, 2024

Agenda

Time	Description	Presenter
11:00- 11:30	Round of Introductions (1min each)	-
11:30-11:45	Project Overview	Ricardo de Casto (UCM)
11:45-12:05	Readiness for EV-based evacuation	Keith Corzine, Osman Saleem (UCSC)
12:05-12:30	Planning Infrastructure Investments	Keith Corzine, Saheed Lakeh (UCSC)
12:30-13:25	Lunch	-
13:25-13:30	Group Photo	-
13:30-14:00	Minimizing Evacuation Time via Optimization Tools	Xuchang Tang (UCD), Shuang Feng (UCM)
14:00-14:20	Definition of use cases for Merced/Mariposa evacuations	Qijian Gan(UCB)
14:20:14:40	Potential Public Policy Implications for effective EV-based evacuations	S. Markolf (UCM)
14:40-15:00	Conclusion & Next Steps	

*10 min presenting + 10 min Q&A

Round of Introductions

1. Name?
2. Affiliation?
3. Experience with evacuations?
(professional or personal)
4. What is your expectation for this meeting?

#	Organization	Name
1	UC Merced	Ricardo de Castro
2	UC Merced	Sam Markolf
3	UC Merced	Shuang Feng
4	UC Merced	Dilawer Ali
5	UC Merced	Sarah Kurtz
6	UC Merced	Farzan ZareAfifi
7	UC Davis	X. Lin
8	UC Davis	<u>Xuchang Tang</u>
9	UC Santa Cruz	Keith Corzine
10	UC Santa Cruz	Osman Saleem
11	UC Santa Cruz	Saeed <u>Lakeh</u>
12	<i>UC Berkeley</i>	Scott Moura
13	UC Berkeley	<u>Qijian Gan</u>
14	<u>Calstart</u>	Valerie Thorsen
15	<i>Merced Office of Emergency Services</i>	Adam Amaral
16	<i>Merced Fire Chief</i>	Derek Parker
17	<i>SJ Air Valley Air Pollution District</i>	Kevin Wing
18	SJ EV Partnership	Ollie Danner
19	Mariposa Office of Emergency Services	Wes
20	Perimeter	Noah Wu
21	UC Merced Office of Emergency	Laura Mascorro

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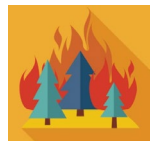
*10 min presenting + 10 Q&A

Natural Hazards Trigger Evacuations in California

2023: Planada (Merced) Enters Third Day of Evacuations
as More **Storms** Loom Ahead ...



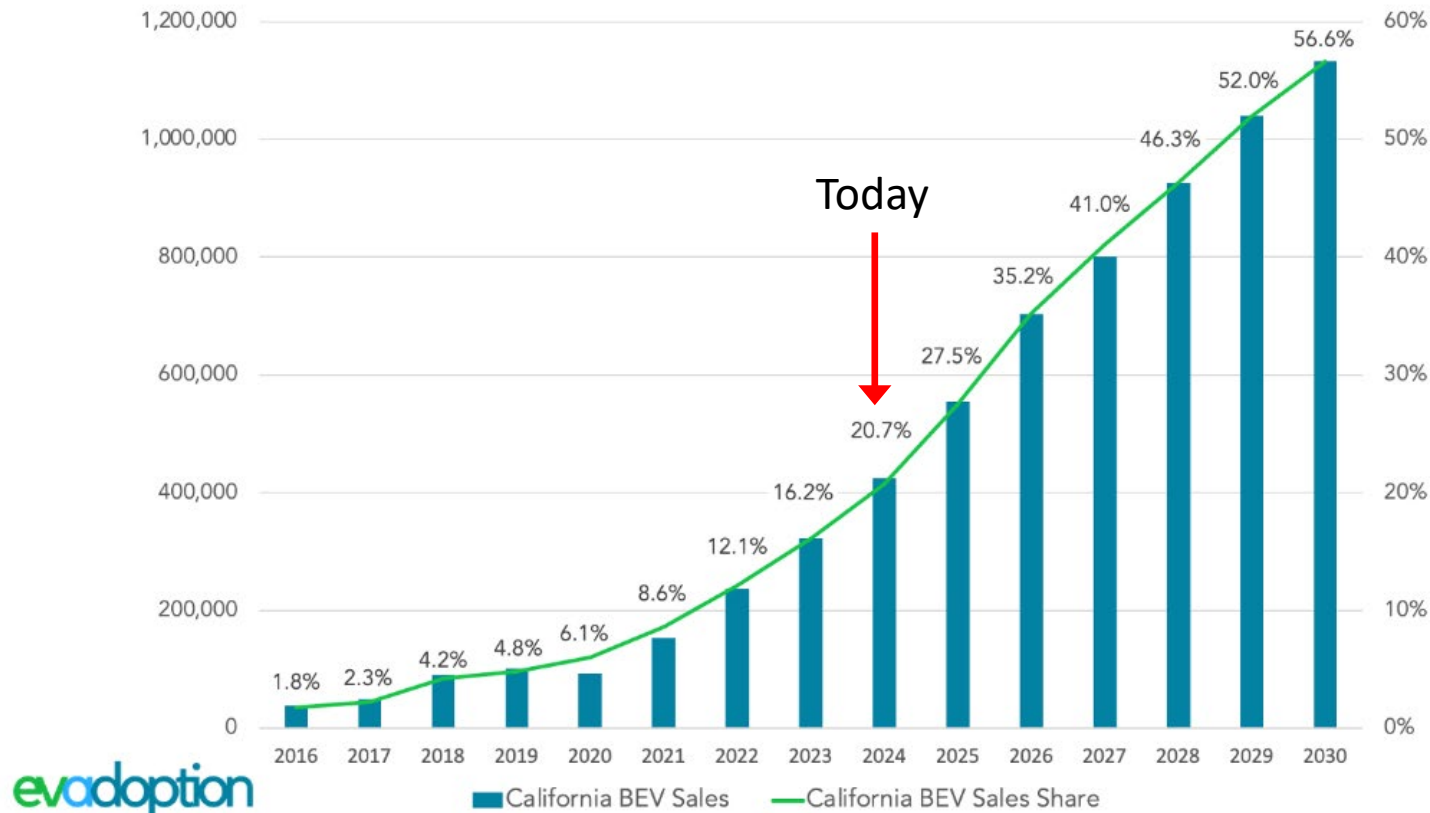
2022: Oak **fire** near Yosemite explores to 6555
acres. Thousands evacuated



2007: mass evacuations (**500,000**
people) over a 7day period in San
Diego due to **Witch Fire**

Mass adoption of EVs in California

California BEV Sales & New Vehicle Share Forecast: 2016-2030



CA Executive order N-79-20:
100% sales of light duty Zero
Emission Vehicles (ZEV) by
2035

**Q: Are we ready to deal with
evacuations using only EVs?
What could be the challenges?**

Challenge 1: Lack of Power

California faces historic power outage due to fire danger

BY ASSOCIATED PRESS

OCTOBER 8, 2019

Share   

SAN FRANCISCO (AP) — Millions of people were poised to lose electricity throughout northern and central California after Pacific Gas & Electric Co. announced Tuesday it would shut off power in the largest preventive outage in state history to try to avert wildfires caused by faulty lines.

1. How to evacuate when **grid power is not available** to charge EVs?

Challenge: Grid Overload

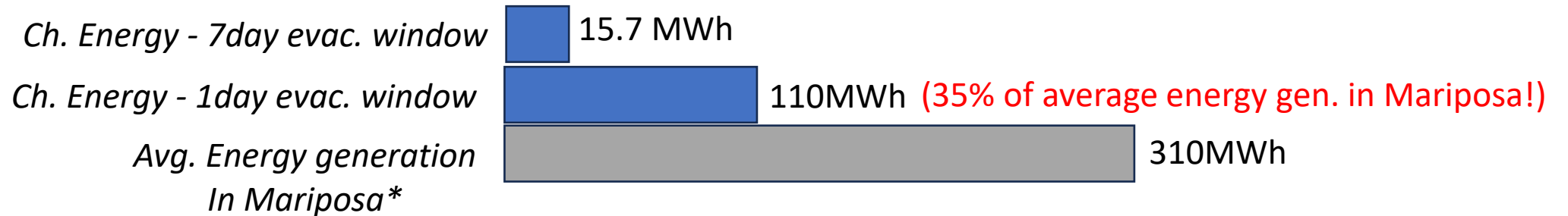
Assumptions

- Evacuation of 5k cars in Mariposa
- 0.32kWh/mile of energy consumption
- 70 miles to reach “safety”/shelter

Evacuation Energy Needs

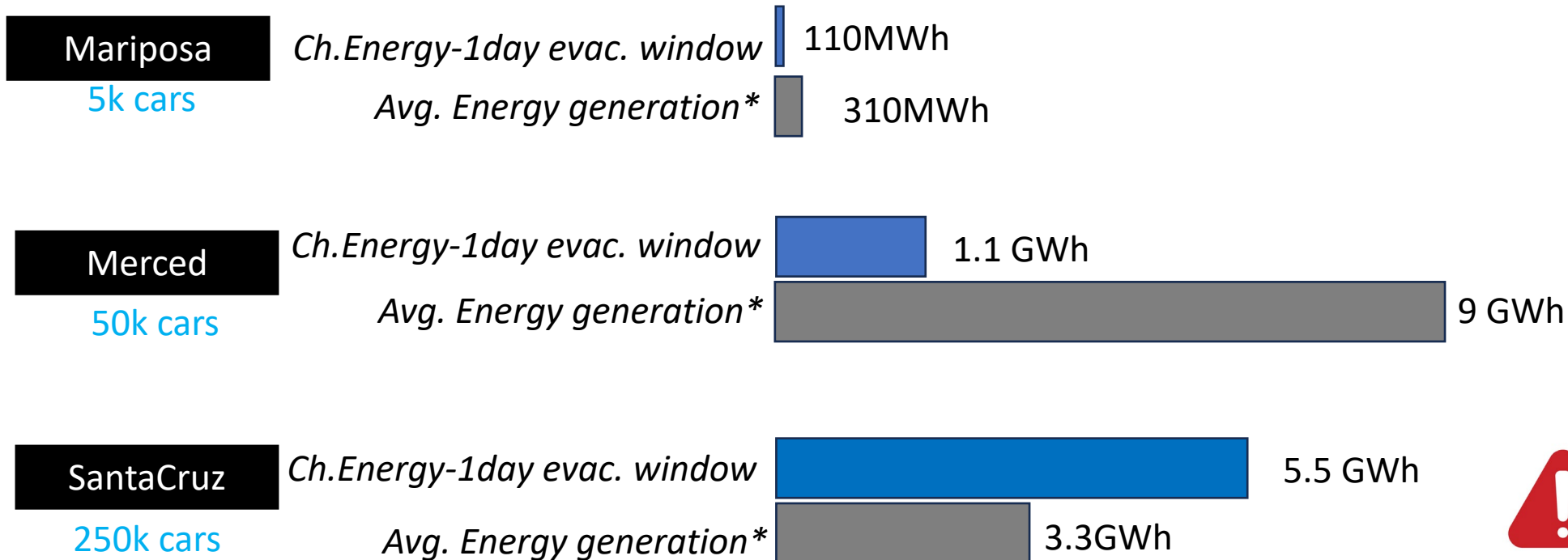
- Energy: 22kWh/vehicle
- Total Evac. Energy: 22kWh*5k = **110MWh**

Short evacuation window may **overload the grid!**



*Mariposa avg power in 2022: 114GWh/365=0.31GWh/day=**13MW**

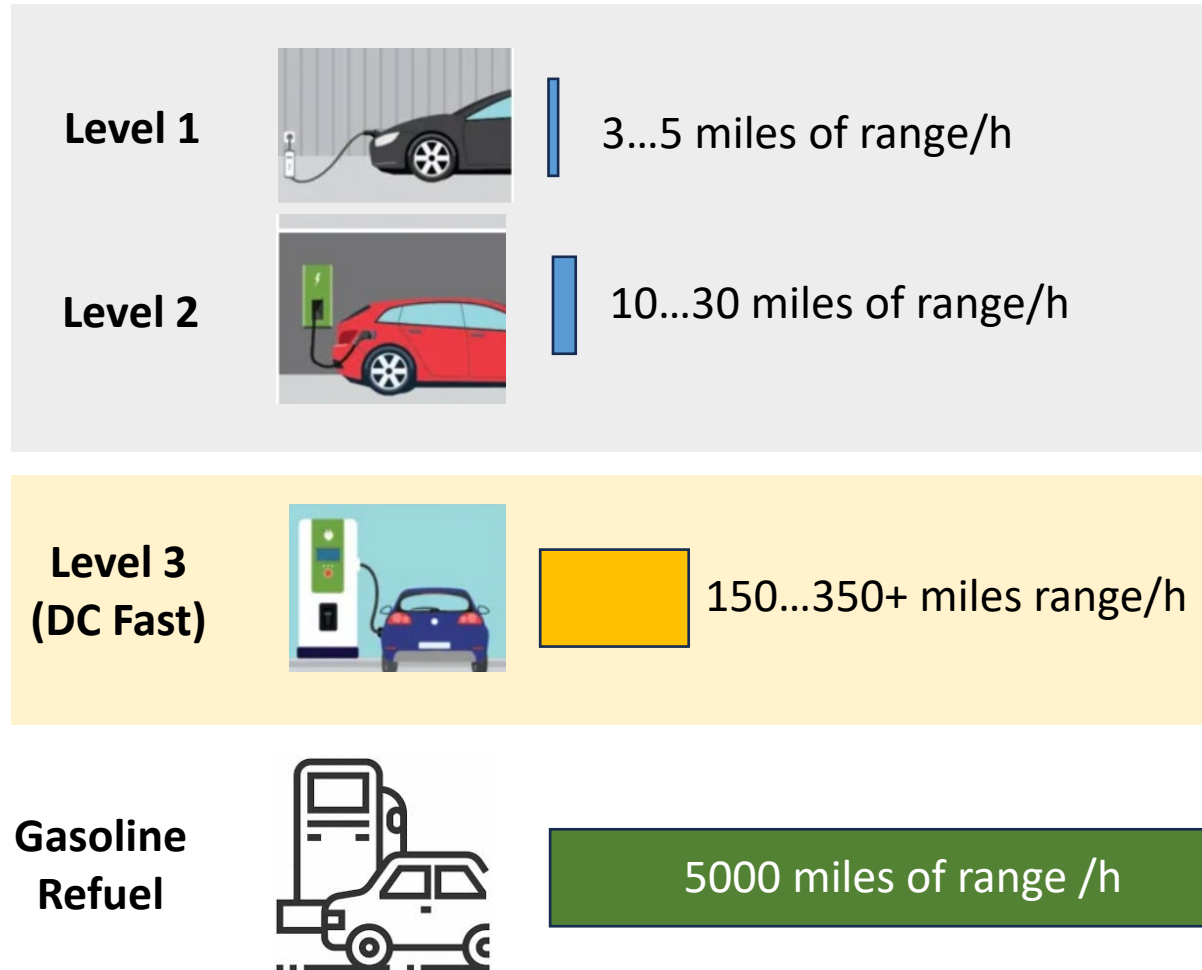
Challenge: Grid Overload



Ch Energy Demand > Avg Energy Generation!

*What if we need to evacuate and the grid is **overloaded**?*

Challenge: Charging Time



→ L1/L2 charge at home is good option if we have long time to evacuate

→ L3 Charging more attractive for “quick” evacuations, but still 10x to 30x slower than gasoline vehicles

3: How to minimize charging time/queuing during evacuation?

Credits: City of Phoenix

Challenge: Public Policy

- “When backup power becomes law” (Fla. Stat. § 526.143)
 - *requires gas stations located along evacuation routes to have backup power to pump fuel so residents can evacuate before a storm*
 - *business must keep a written statement attesting to the periodic testing and ensured operational capacity of the equipment.*
 - *required documents must be made available, upon request, to the county emergency management office*

4: Should we mandate Backup power in EV charging stations that are located along critical evacuation routes?

Source: Hurricane Irma’s Effect on Florida’s Fuel Distribution System and Recommended Improvements

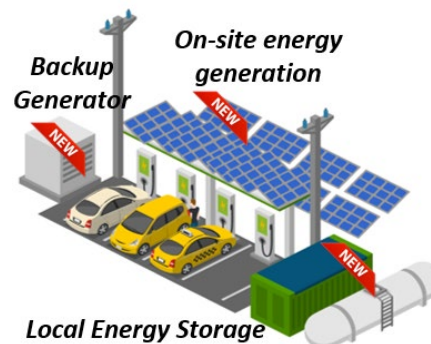
Project Outcomes/Objectives

O1. A new ZEV Evacuation Readiness Score

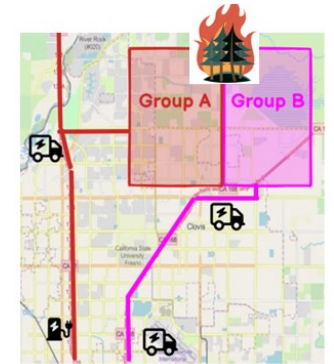


O2/O3: Decision Aid Tools

O2. planning investments and reduce vulnerabilities



O3. Evacuation plans (route, charging scheduling)



O4. Policy Recommendations



White paper to support effective ZEV evacuations

O5. Resources/Education Materials/Outreach

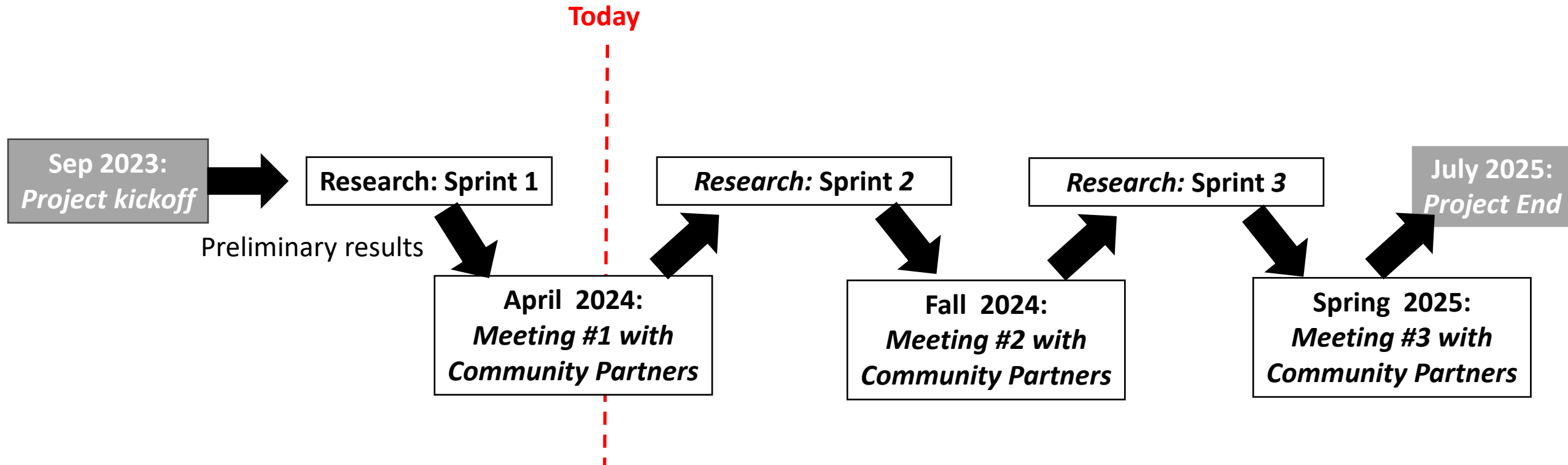


Factsheets, checklists, flowcharts, workshops

Team & Community Partners



Project Timeline



Share with community partners progress of the research team

Objectives for the Day

- Share with community partners progress of the research team
- Collect feedback from partners. Some questions:
 - Do you think the goals are sound and have practical relevance?
 - Are you aware of other EV challenges during evacuations that we overlooked?
 - We would like to build evacuation use cases for Central Valley/Sierra Nevada/Sacramento/Bay Area/Santa Cruz. Are you available to help us create these use cases?

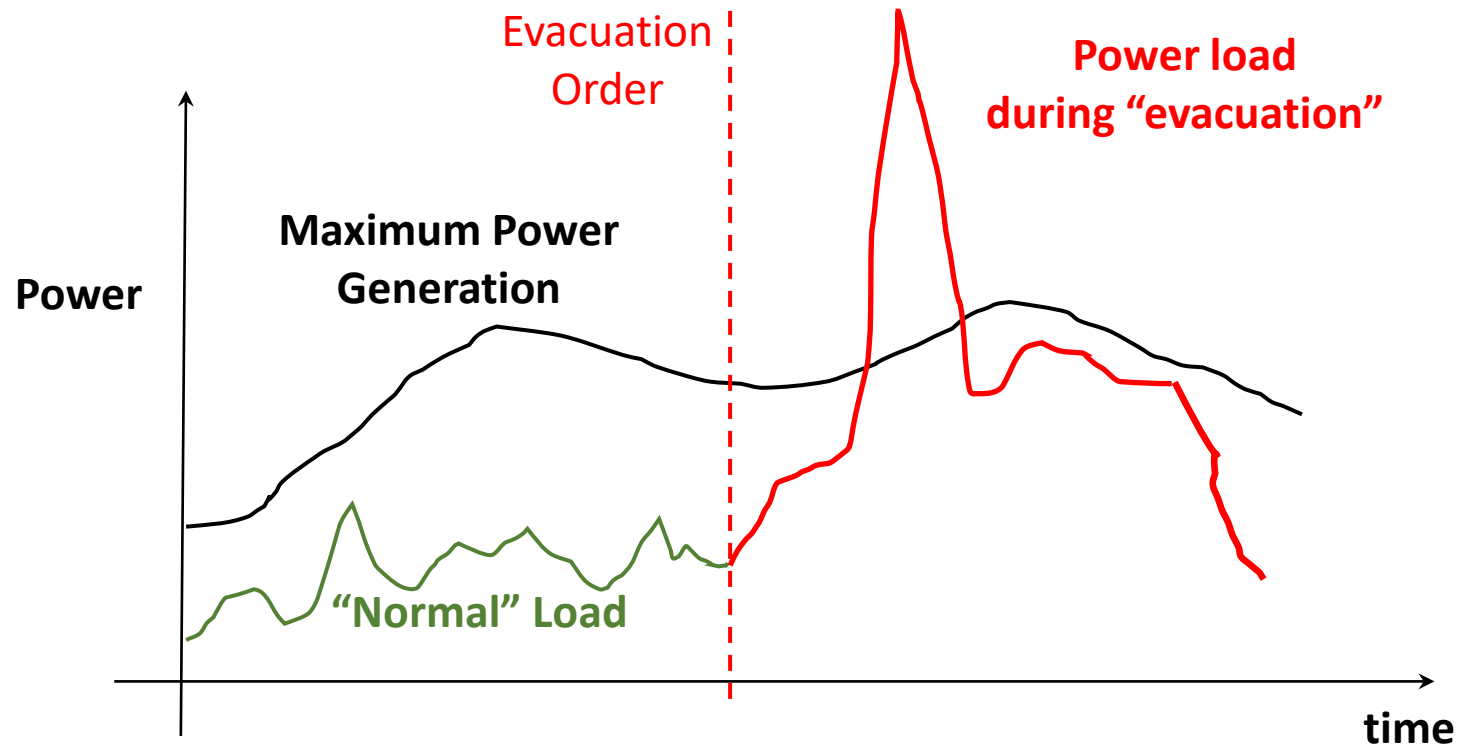
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- Extra Slides

Challenge 2: Grid Overloading



2. What if we need to evacuate and the grid is **overloaded**?

Improving Preparedness of Communities for Evacuations using ZEVs

Development of ZEV Evacuation Readiness Score

Presenter: Osman Saleem

April 26th, 2024

Background

- What comes to mind when we tend to compare two things?
- Rating/Ranking is the go-to method to rank two products of same kind:



Iphone 15 pro max

Rating: ★★★★★

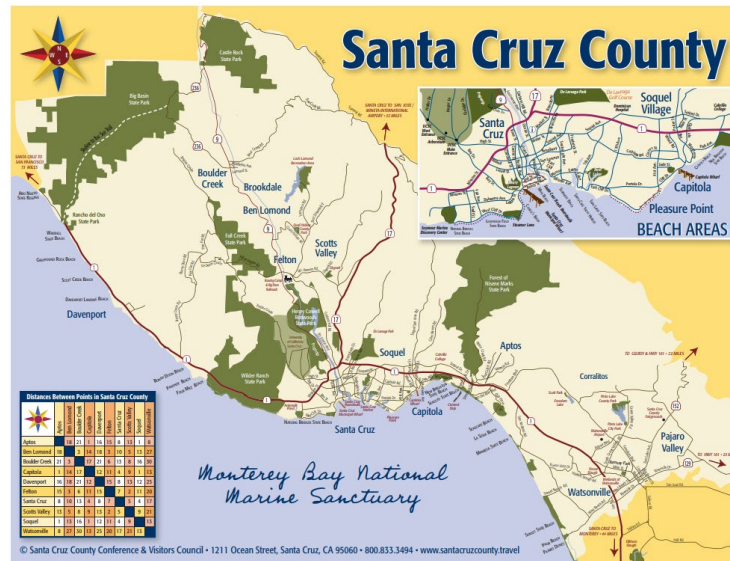


Samsung S24 ultra

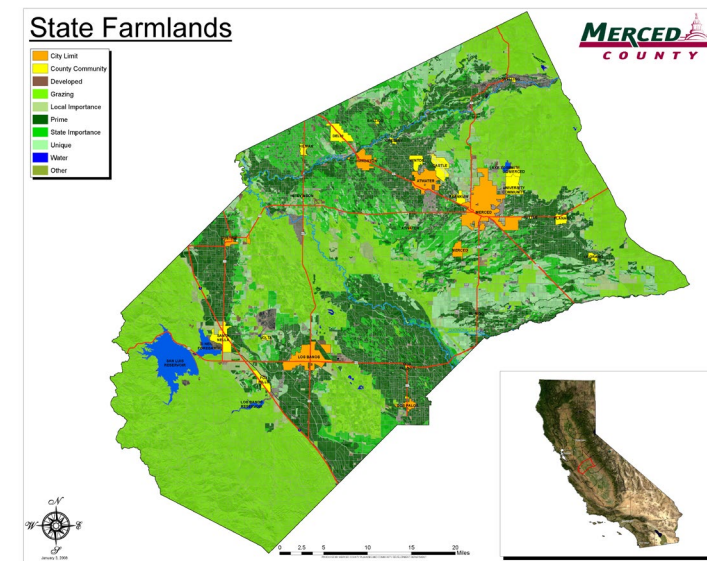
Rating: ★★★★★

Introduction to ZEV Evacuation Score

- ZEV evacuation readiness score provides a basis to categorize each county based on its preparedness to successfully carry out a zero-emission vehicle evacuation.



ZEV Score: ★★★★★

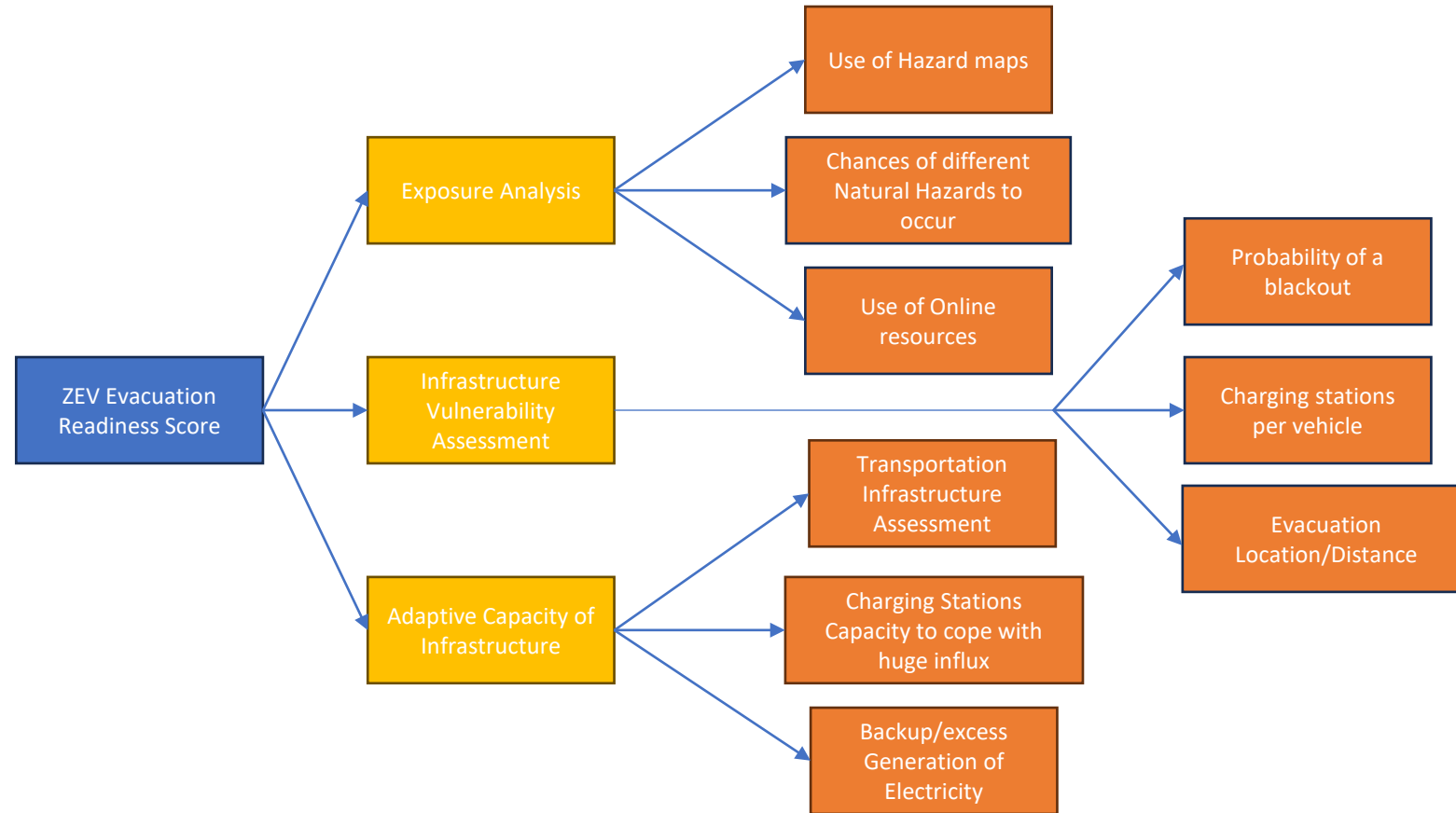


ZEV Score: ★★★★★

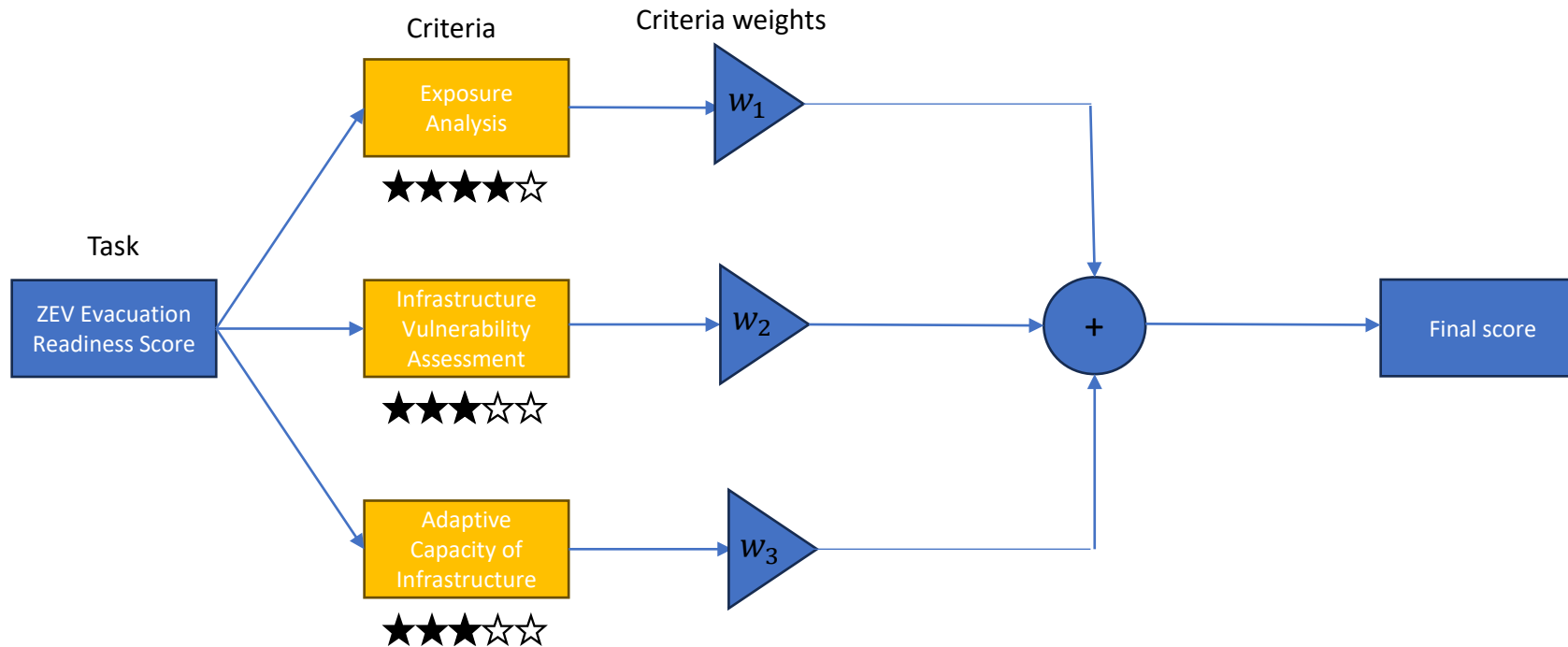
Importance of ZEV Evacuation Score

- This score can help us identify the indicators that need more research and development in a certain county.
- It can also aid public organizations in directing technical and financial support to communities that need more help.
- As this score depends upon different indicators. Certain score can be broken down into its specific indicator indicating which requires more development and funding.

Indicators used in ZEV Evacuation Score



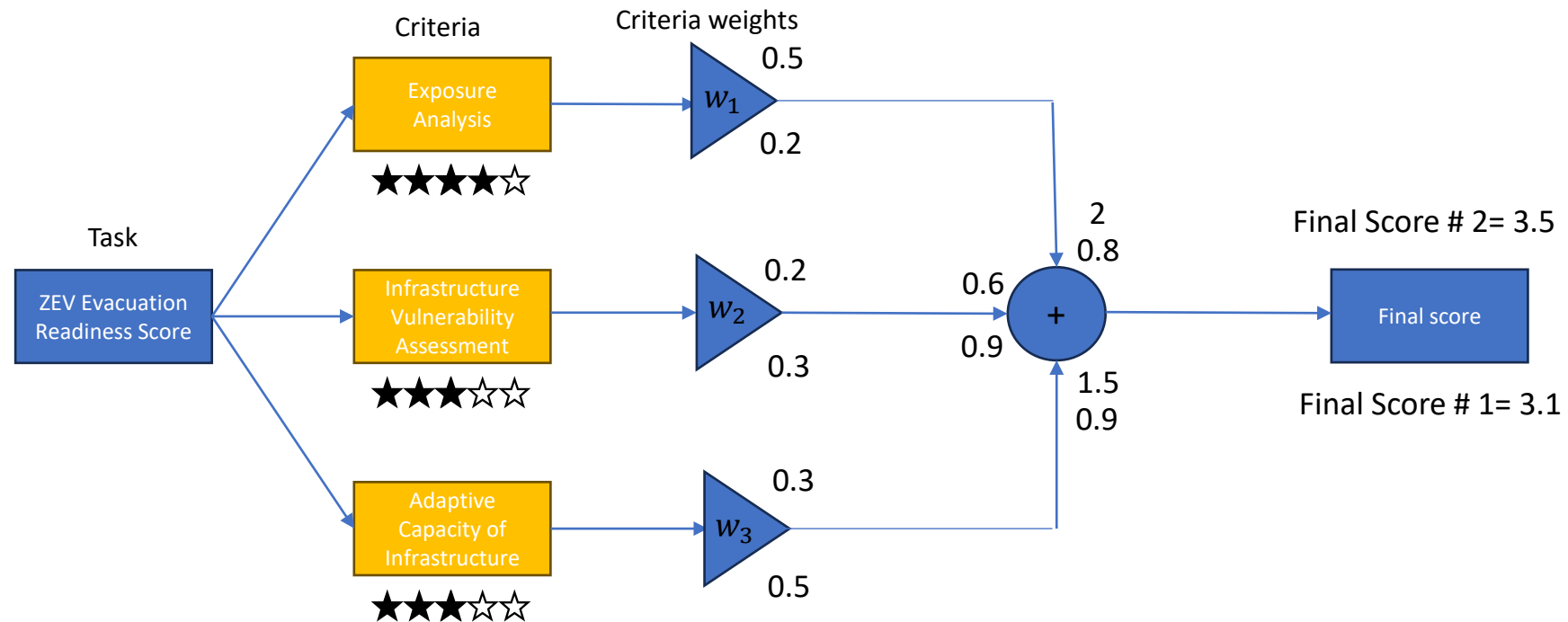
Development of ZEV Score From Indicators



- The weights in the above flowchart depends on the survey result from the community and experts (YOU!).
- How would it affect the score? Let's see!

Importance of Criteria Weights

- Let's calculate ZEV Score for different weights:

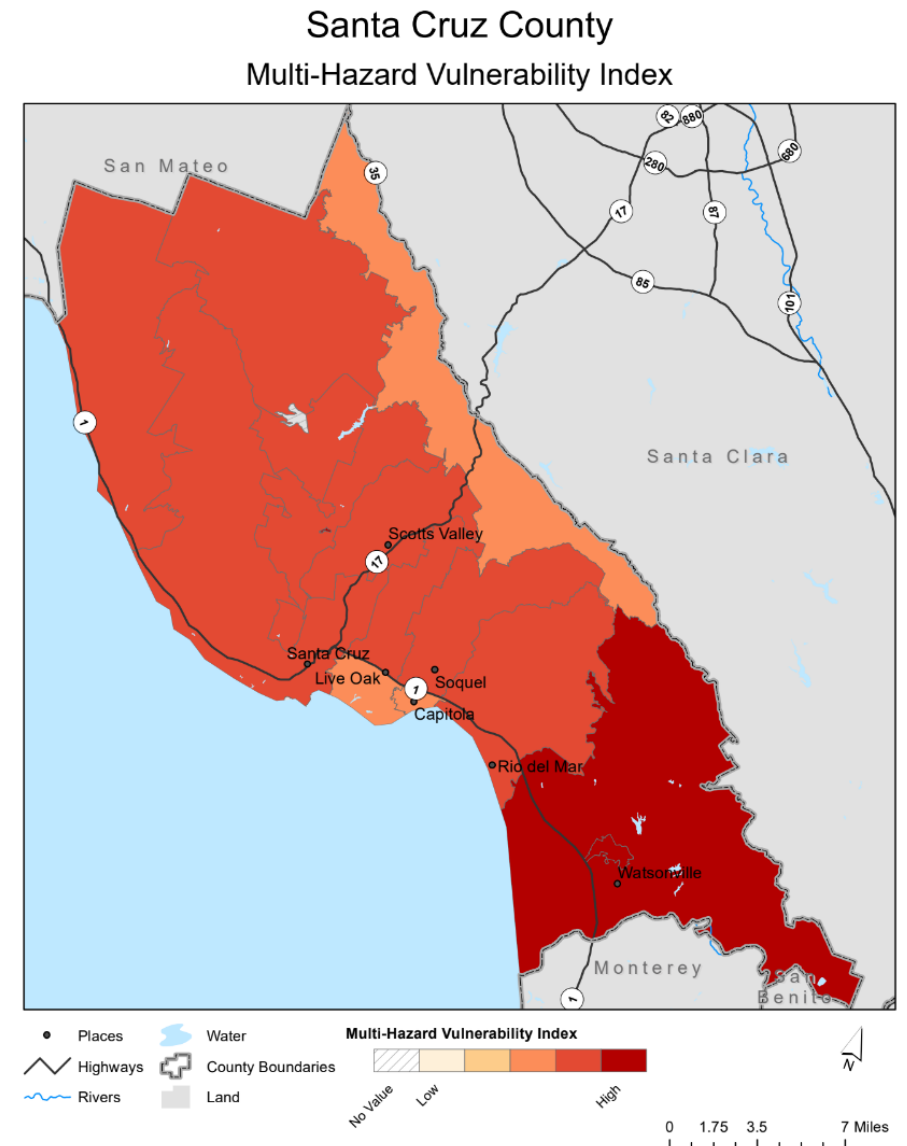


- Let's see what different criteria are in detail.

Exposure Analysis (EI)

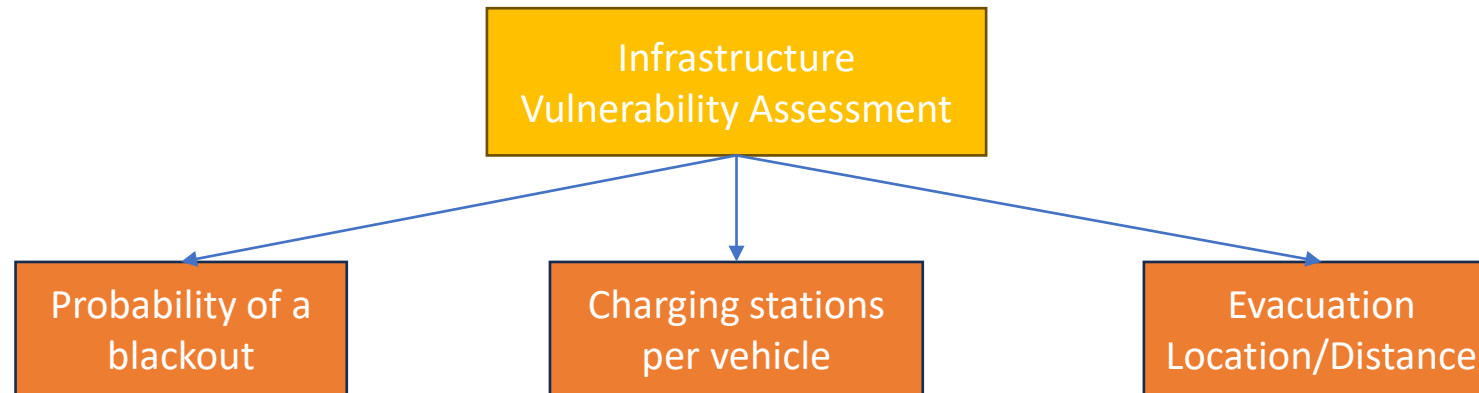
- This criterion deals with the estimation of the level to which the area/county is prone to certain hazards.
- These hazards can include wildfires, floods, earthquakes, or an overall estimation of all kinds of hazards.
- In our study, we are giving a score to each county based on maps developed for hazard analysis and estimation methods.

Exposure Index: 0.8872975397361342



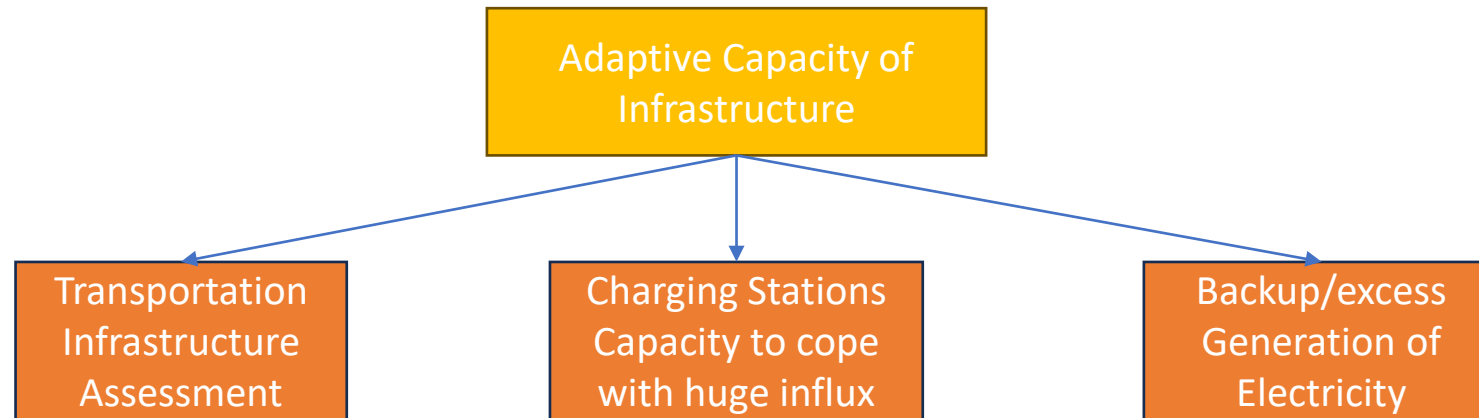
Infrastructure Vulnerability Assessment (IVS)

- This criterion deals with the assessment of already available infrastructure.
- This infrastructure include Electric chargers, roads, electricity transmission network etc.
- We are assessing the ability of area/county to deal with any natural hazards with already built/developed infrastructure

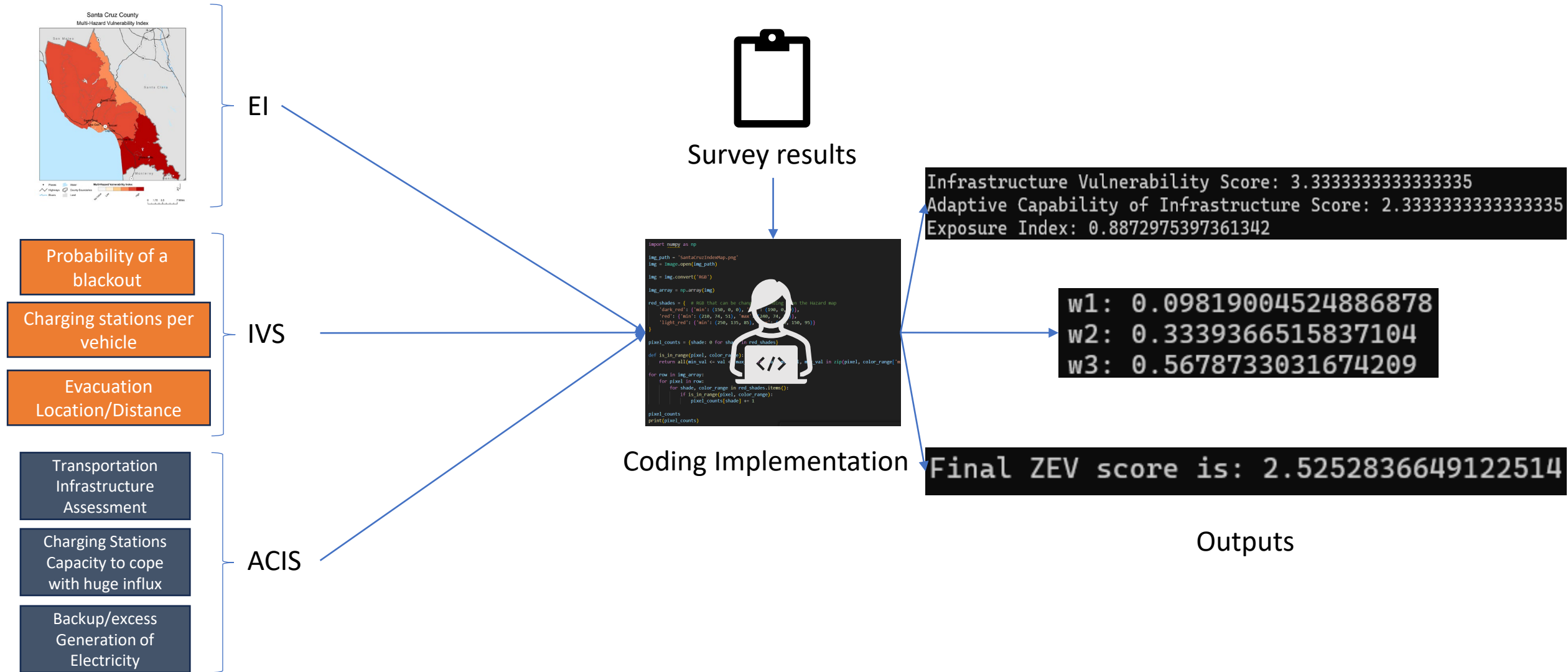


Adaptive Capacity of Infrastructure (ACIS)

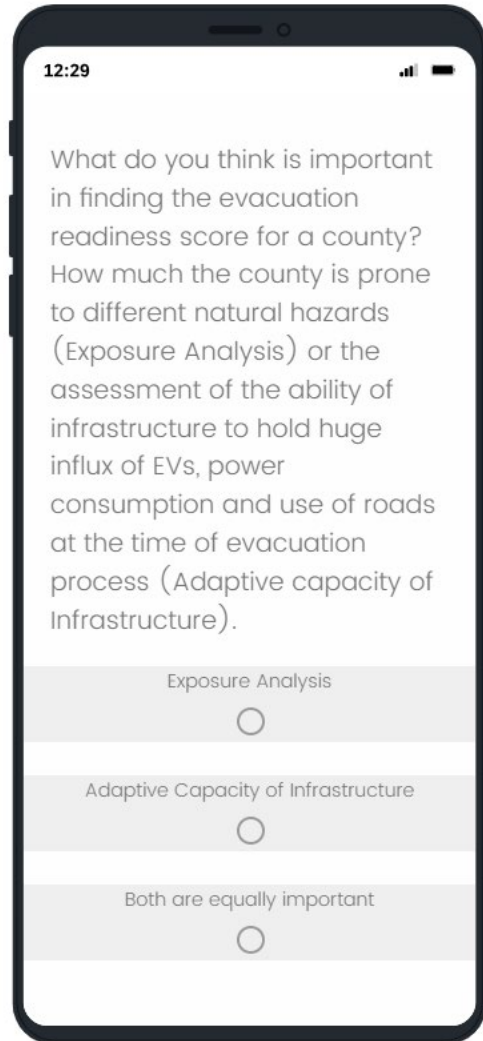
- This criterion deals with the assessment of how adaptive the already available infrastructure is in case of any natural hazard.
- In this criterion, we estimate the ability of electric chargers, power generation units, power transmission systems and roads to take up load and stress in case of evacuation.



Development of a Coding Algorithm



Survey's Overview



12:29

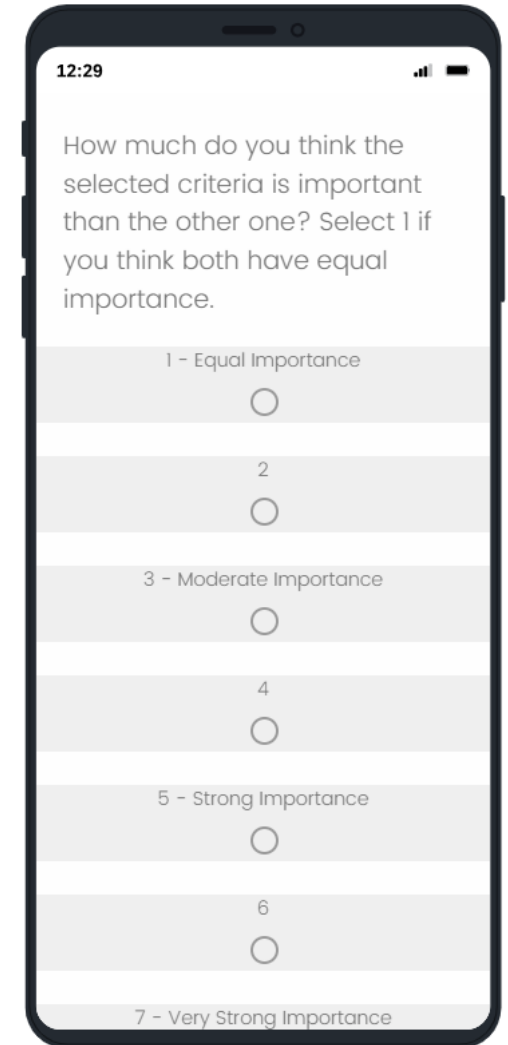
What do you think is important in finding the evacuation readiness score for a county? How much the county is prone to different natural hazards (Exposure Analysis) or the assessment of the ability of infrastructure to hold huge influx of EVs, power consumption and use of roads at the time of evacuation process (Adaptive capacity of Infrastructure).

Exposure Analysis ☐

Adaptive Capacity of Infrastructure ☐

Both are equally important ☐

- There will be 6 questions in total (Estimated time to complete survey: 8-10 mins)
- 1st question deals with the relative importance of different criteria.
- 2nd question delves into deeper comparison and asks for a rating signifying the importance of selected criterion as compared to the other one.



12:29

How much do you think the selected criteria is important than the other one? Select 1 if you think both have equal importance.

1 - Equal Importance ☐

2 ☐

3 - Moderate Importance ☐

4 ☐

5 - Strong Importance ☐

6 ☐

7 - Very Strong Importance ☐

Survey's QR Code



Thank you!



Improving Preparedness of Communities for Evacuations using ZEVs

Task 3: Minimizing Evacuation time via Optimization Tools

Presenter:

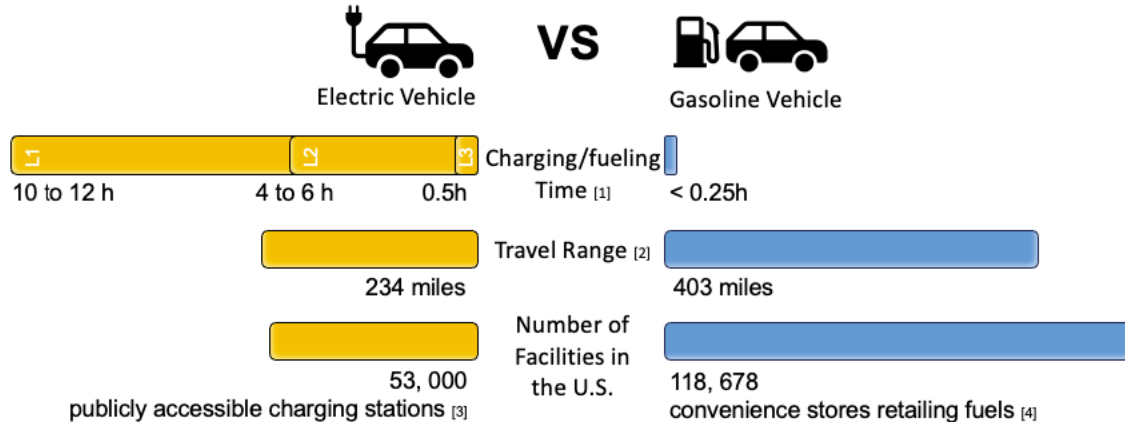
Xuchang Tang (UC Davis)

Shuang Feng (UC Merced)

April 26th, 2024

Introduction

- The objective → **safe** and **efficient** evacuation.
- EVs' limited range and recharging needs pose challenges.
- Insufficient, daily-use oriented, or vulnerable charging infrastructure exacerbates these challenges.



[1] Wildfire Evacuees Fill Lake Tahoe Roads in Rush to Flee

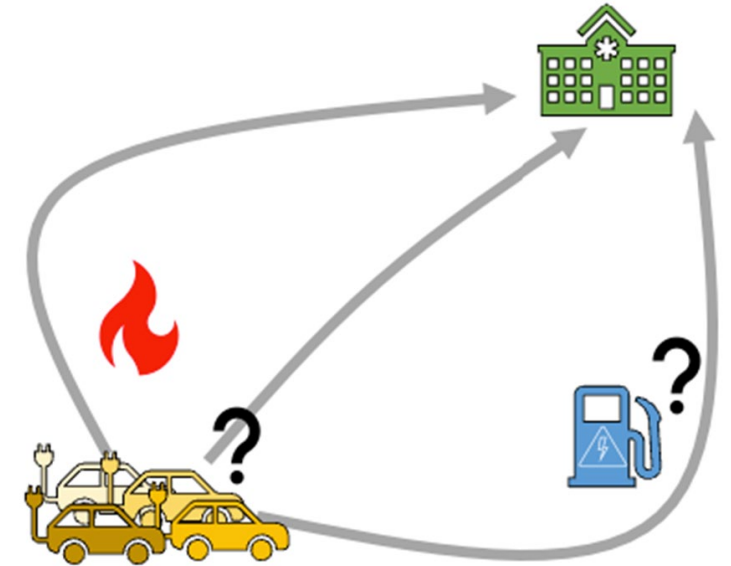


[2] EVs Left Powerless During Hurricanes

Considerations

Goal: **minimize evacuation time**

- Routing
- Charging decisions
- Departure schedule
- Traffic
- Power grid capacity
- Power grid stability
- Mobile charging station (MCS) deployment



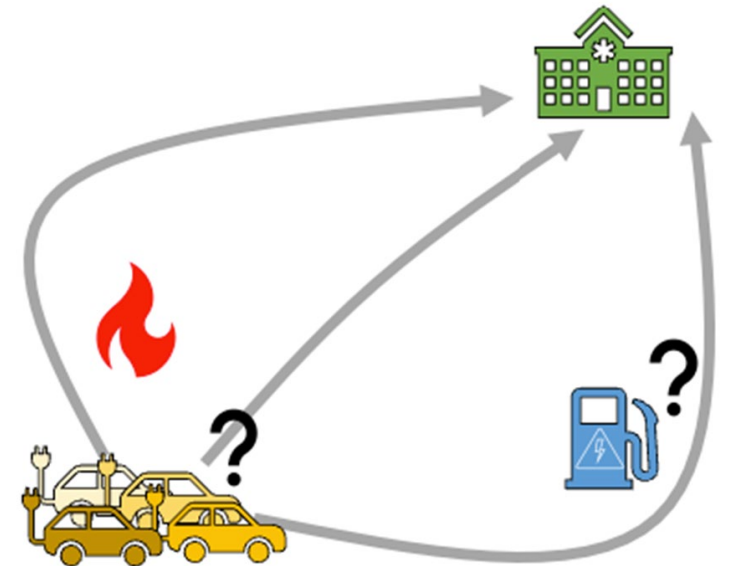
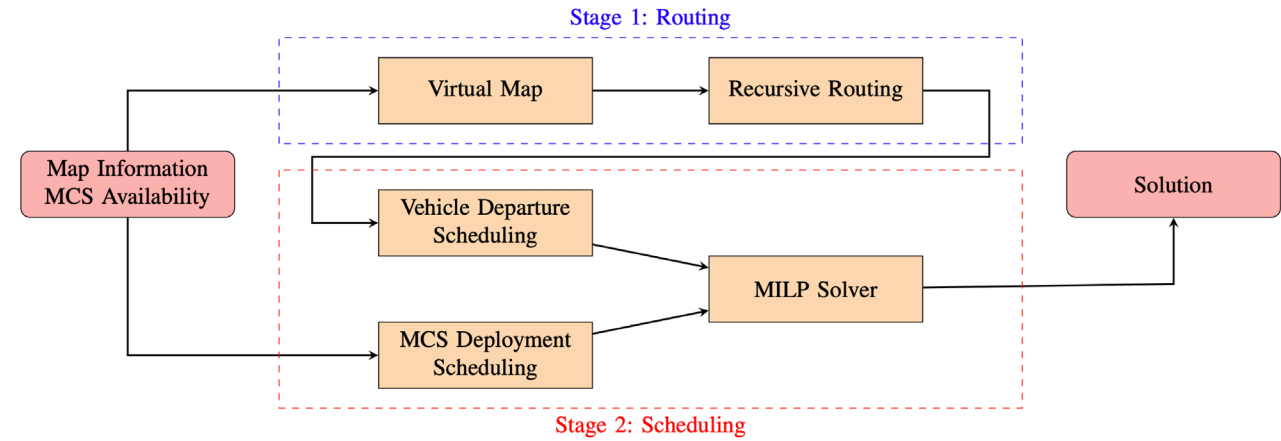
Almost infeasible to solve everything in one step -> Decomposition

Approach #1: Routing Detached

Approach #1: Considerations

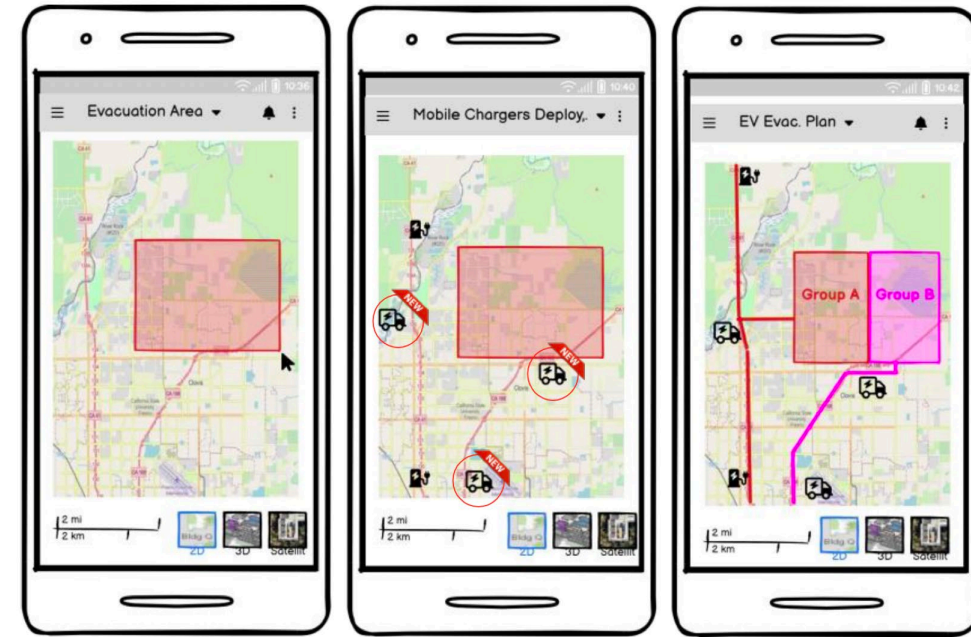
Goal: **minimize evacuation time**

- Routing ✓
- Charging decisions ✓
- Departure schedule ✓
- Traffic
- Power grid capacity ✓ (In terms of ports)
- Power grid stability
- Mobile charging station (MCS) deployment ✓



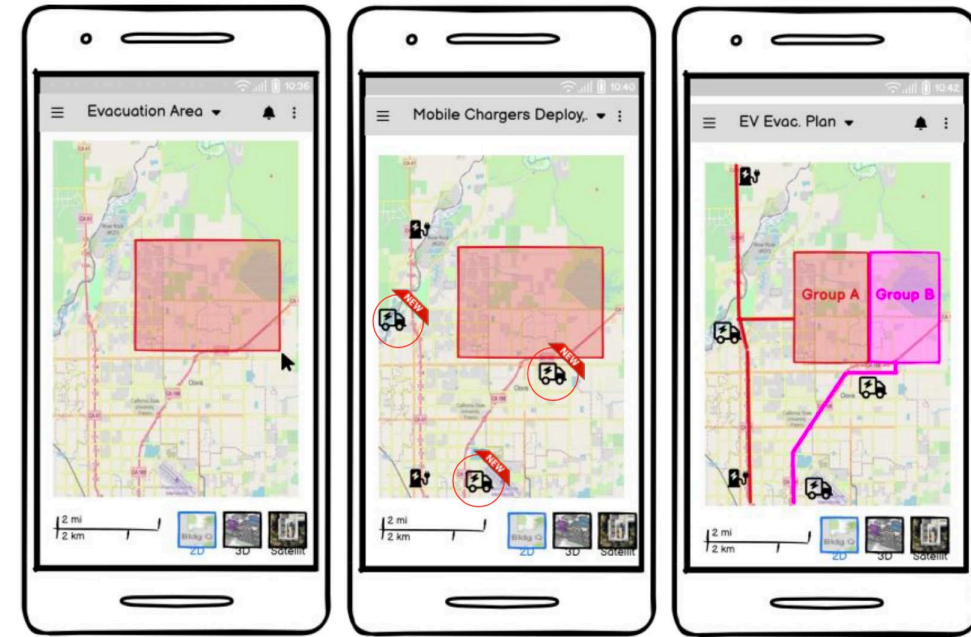
Approach #1: Interactive Tool (Input)

- Map (from **authority**)
 - Connectivity
 - Distances
 - Road conditions (travel efficiency & speed limit)
 - Pre-determined evacuation routes (if applicable)
 - Charging station information
 - Location
 - Number of ports
 - Charging rate
 - Maximum wattage



Approach #1: Interactive Tool (Input)

- Evacuation Demand
 - Origin & Destination (from **user**)
- Hyperparameters
 - Vehicle type (from **user**)
 - At-home state of charge (SOC) (from **user**)
 - Evacuation time requirement (from **authority**)
 - Mobile charging stations (MCS) availabilities (from **authority**)



Approach #1: Optimization Math Engine

Mixed Integer Linear Programming Formulation

- Decision Variables

- x_{wpt} : Number of vehicle evacuated between origin-destination (OD) pair w following path p at time t
- q_{mnt} : Binary deployment status of MCS labeled m at node n at time t

- Objective:

- Summed evacuation time:
$$\sum_{w \in W} \sum_{p \in P_w} \sum_{t \in T} (t + \tilde{t}_{wpt}) \times x_{wpt}$$

- Constraints

- Evacuation demand:
$$\sum_{p \in P_w} \sum_{t \in T} x_{wpt} = f_w, \forall w \in W$$
- MCS single site deployment:
$$\sum |q_{mnt}| = 1, \forall m \in M, \forall t \in T$$
- Port limit: $D_{nt} \leq S_{nt}, \forall n \in N, \forall t \in T$
- MCS energy limit: $0 \leq soc_{mt} \leq 1, \forall m \in M, \forall t \in T$

Approach #1: Optimization Math Engine

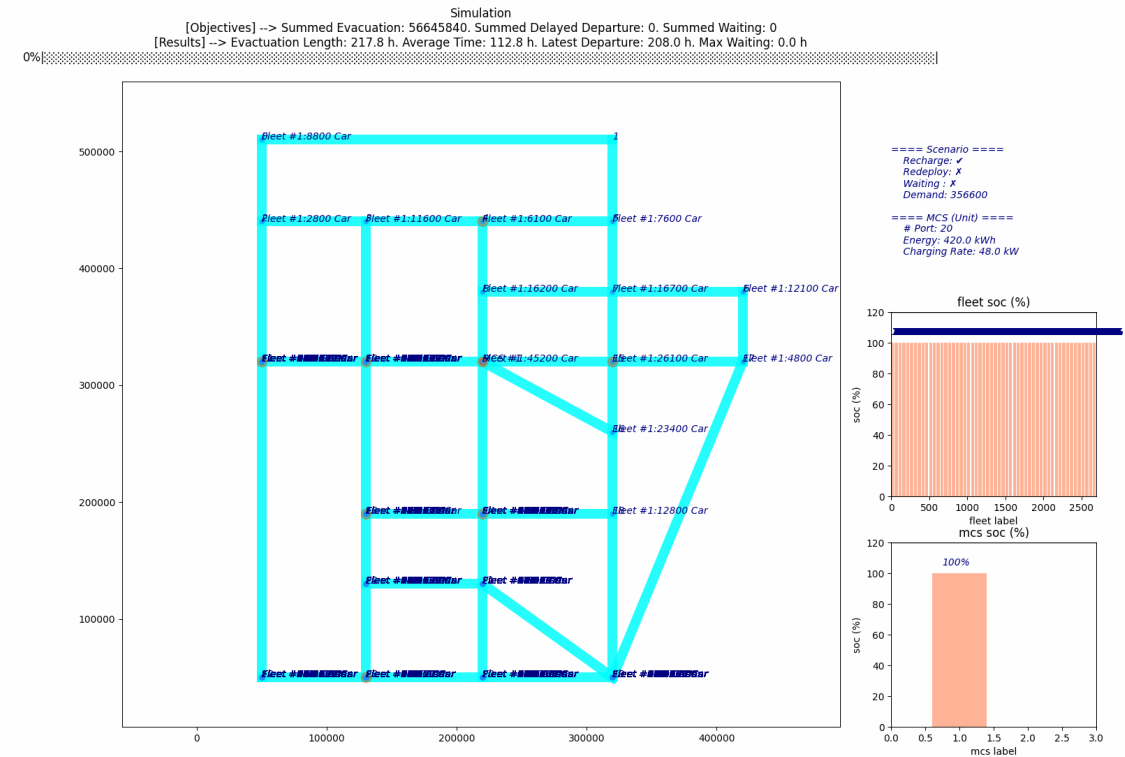
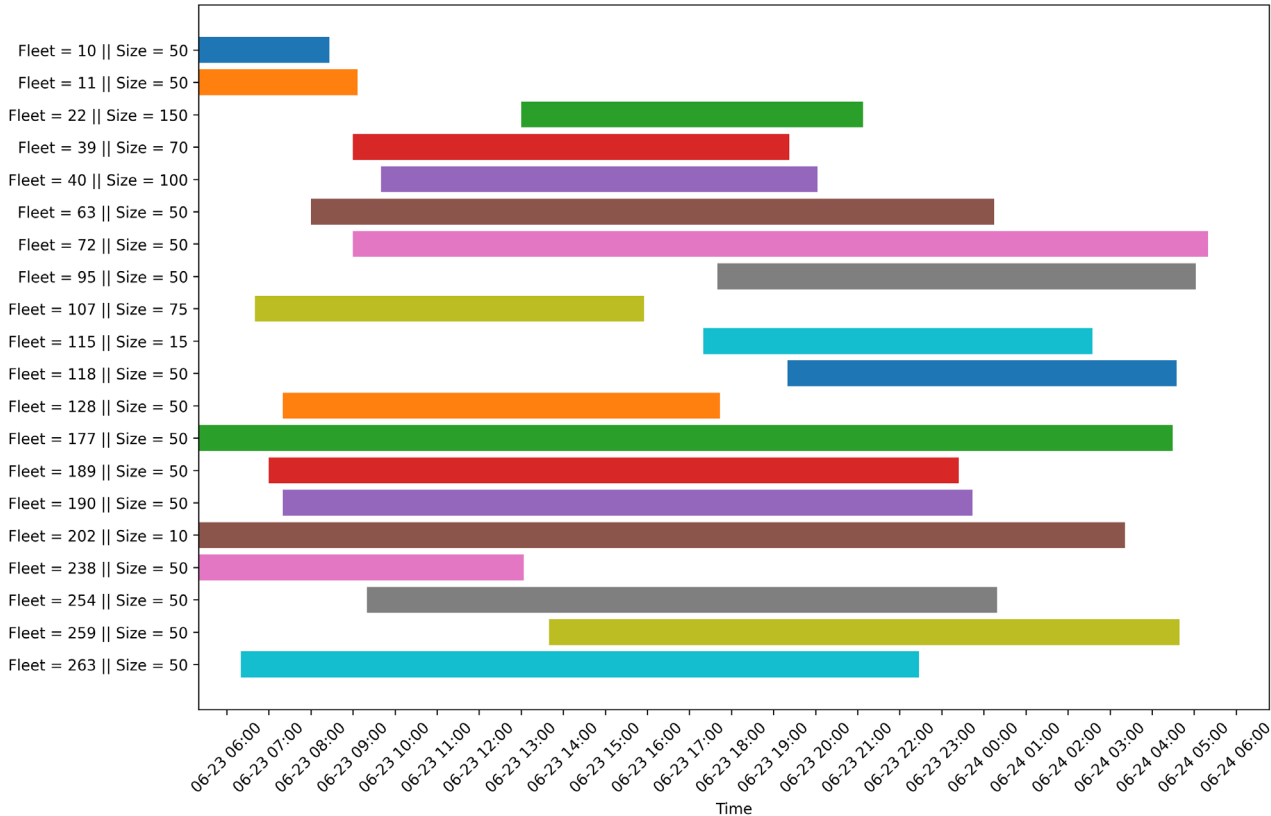
Mixed Integer Linear Programming Formulation

- Objective —> Minimize the **summed** evacuation time
- Constraints:
 - Fit-the-demand —> **All** registered **vehicles** have to be **evacuated**
 - Pre-deployment —> **MCS** are placed at a **single site** throughout the evacuation
 - Port-limit —> **charging demands** at sites are **bounded** by number of **ports**
 - Energy-limit —> **MCS SOC** is **bounded** between 0 and 1

Approach #1: Results

Gantt chart & Animation

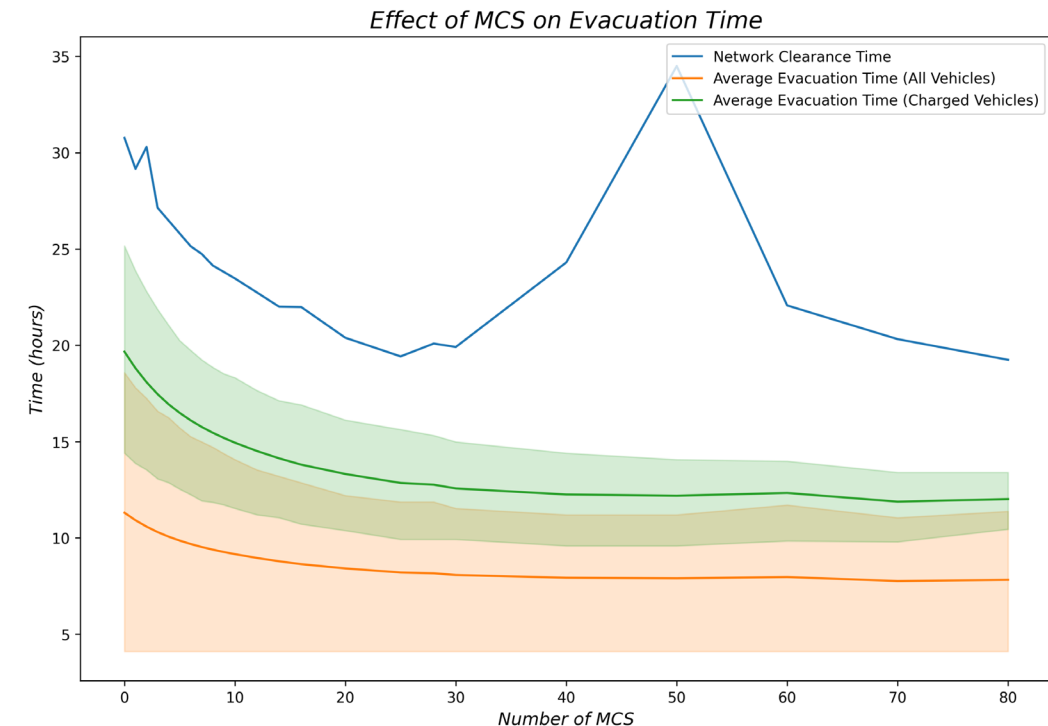
Example Evacuation Gantt Chart



Approach #1: Observations

Reduction of average evacuation time

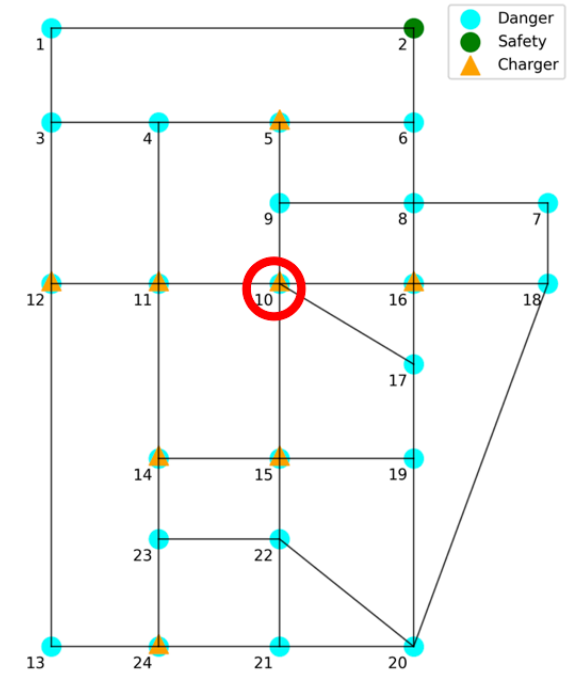
- Shaded = 25% and 75% quantile
- More MCS = Reduced time
- **Diminishing** effect
- Low average faster evacuation span
 - Question for focus:
 - Average?
 - Longest?



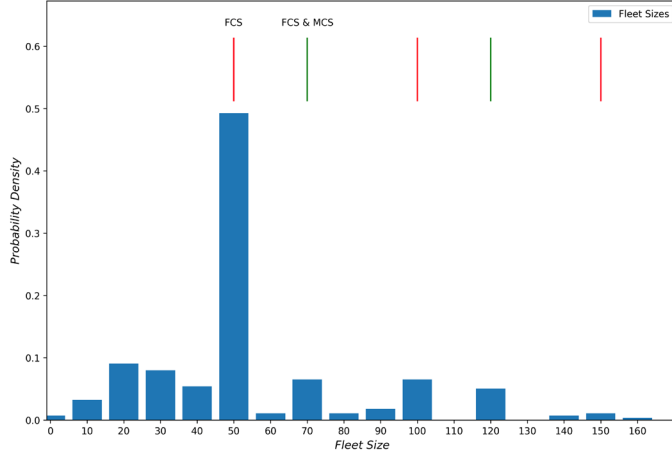
Approach #1: Observations

Optimal scheduling strategy

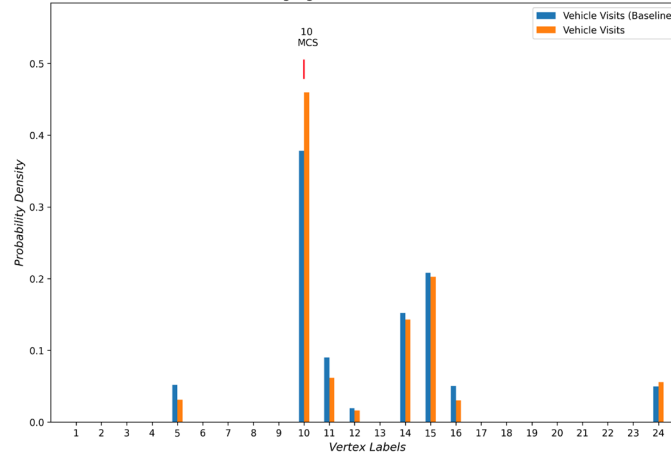
- Vehicle departure
 - Group (fleet) **size** distribution **peaks** at number of **ports**
- MCS placement
 - All placed at the the **most-visited** node
 - The node also has **highest centrality** measures



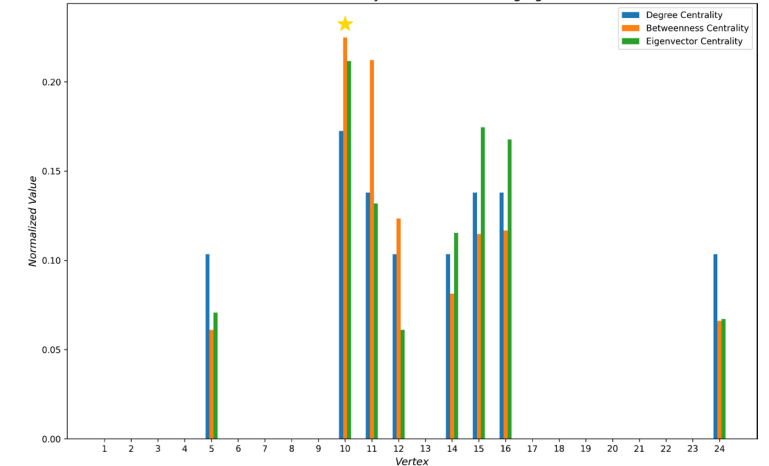
Distribution of Fleet Sizes



Vehicle Charging Activities and MCS Placement

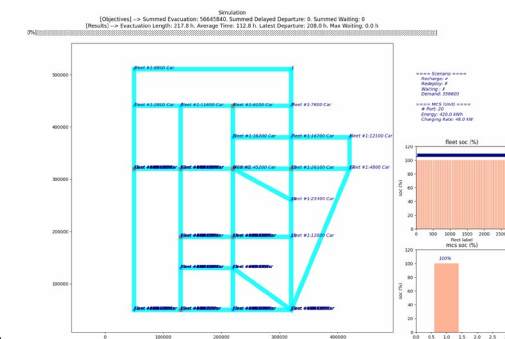
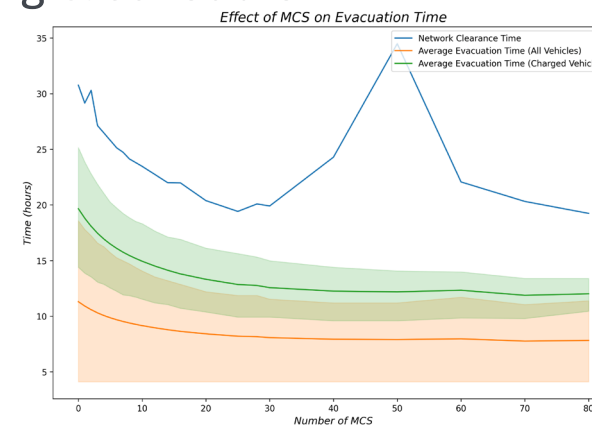
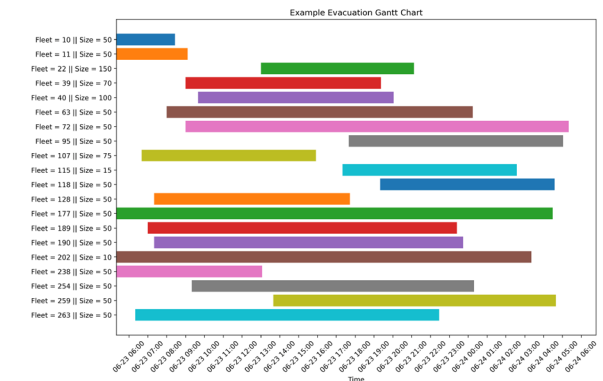
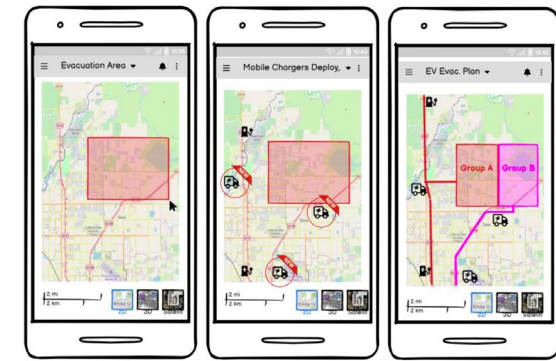
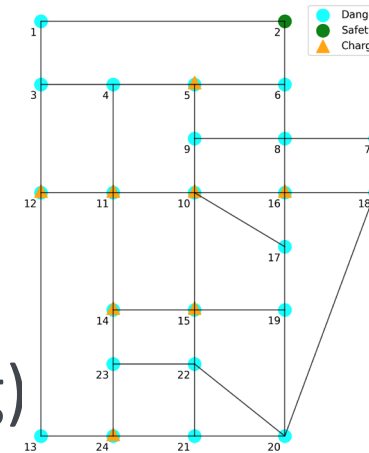


Normalized Centrality Measures for Charging Nodes



Approach #1: Conclusion

- Algorithm (at-heart of evacuation planning)
 - Input:
 - Map (from authority)
 - Hyperparameters (from both users and authority)
 - Evacuation demand (from users)
 - Output:
 - Coordinated vehicle departure grouping & schedule
 - Optimal MCS placement decision
 - Expected outcome:
 - Reduced average evacuation time

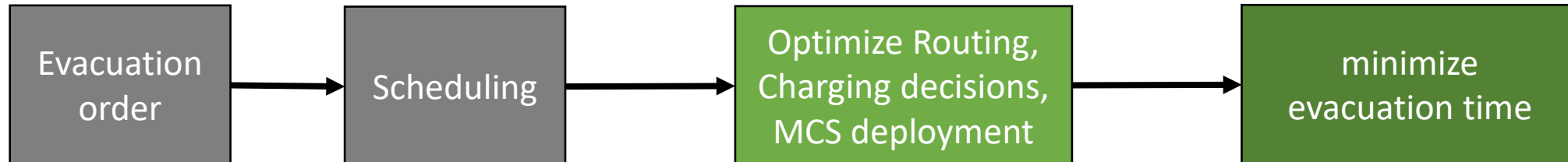
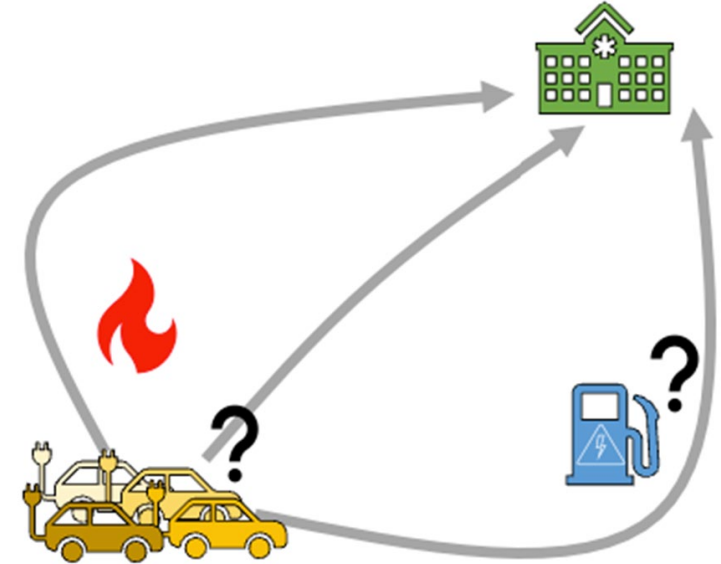


Approach #2: Scheduling Detached

Approach #2: Considerations

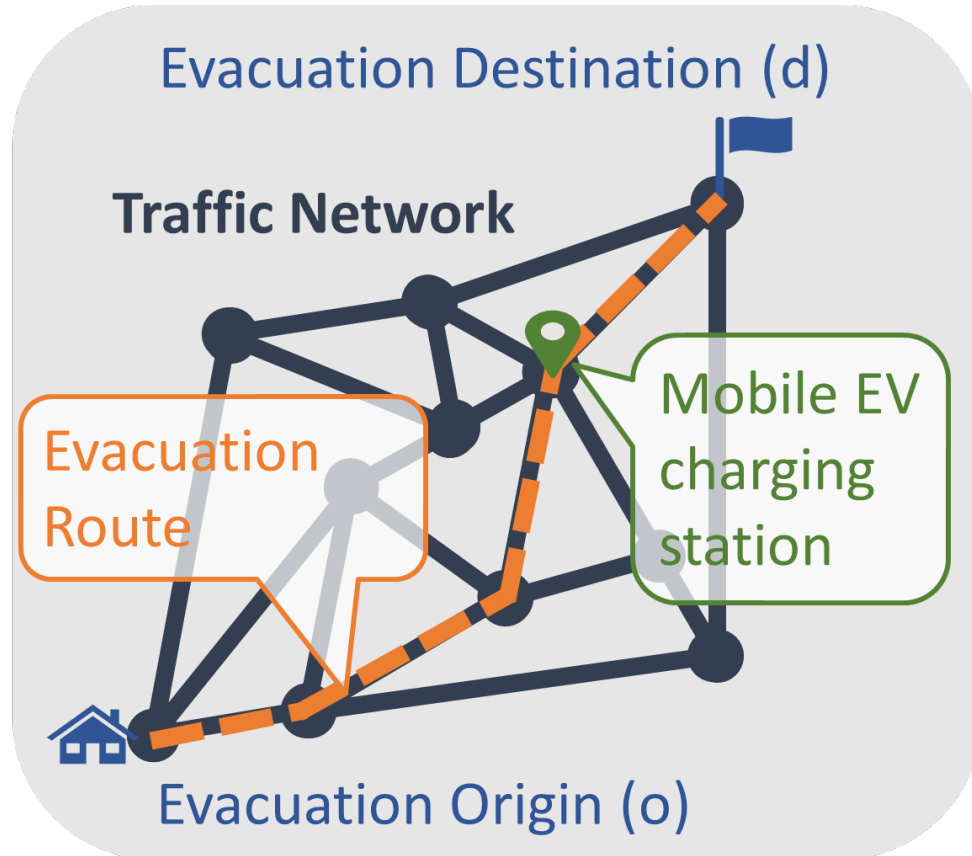
Goal: **minimize evacuation time**

- Routing ✓
- Charging decisions ✓
- Departure schedule
- Traffic ✓
- Power grid capacity
- Power grid stability
- Mobile charging station (MCS) deployment ✓



Approach #2: Methods

- **Network Facility-Location-Routing Problem (FLRP) [1]**



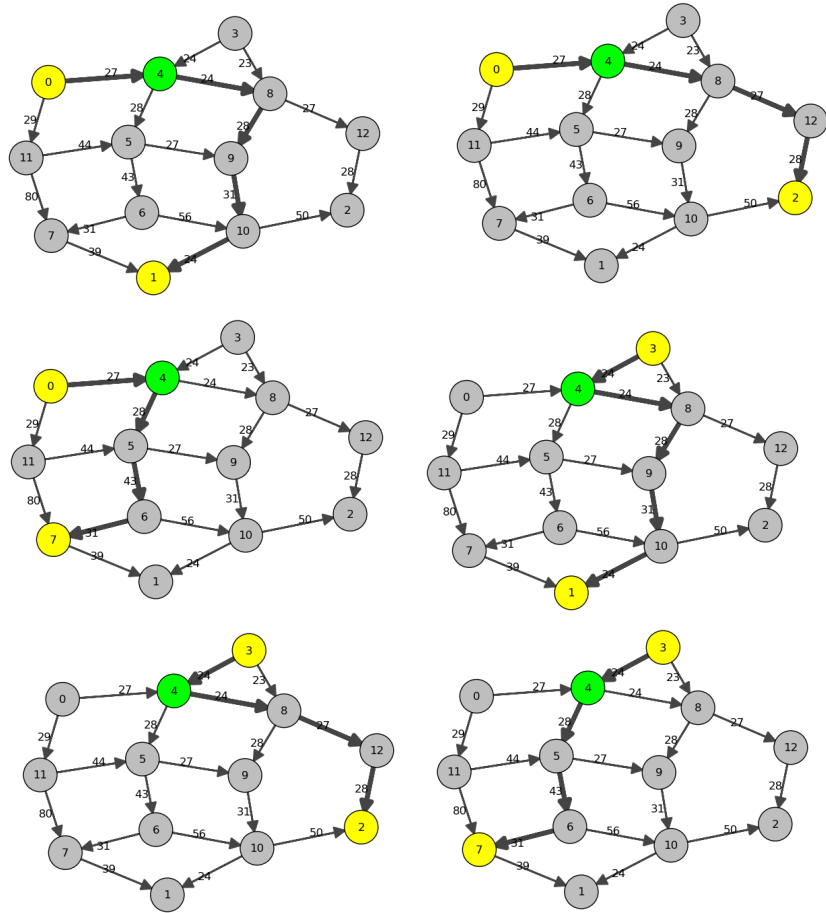
Objective:

- Minimize the **evacuation time** (travel time + charging time + number of stops made during evacuation) and the **number of mobile charging stations placed** for evacuation

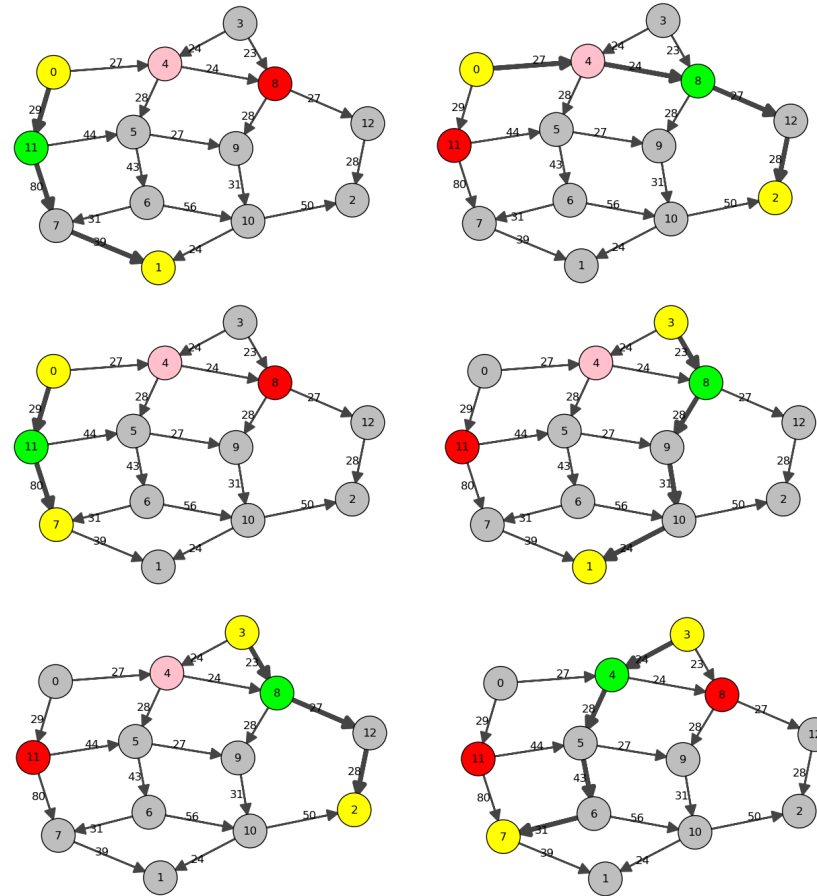
Constraints:

- All EVs can **safely** arrived at evacuation destination without running out of battery.
- Not overcharge EV batteries.
- **More traffic** on the road -> **longer traveling time** (Bureau of Public Roads (BPR) function)
- ...

Approach #2: Results



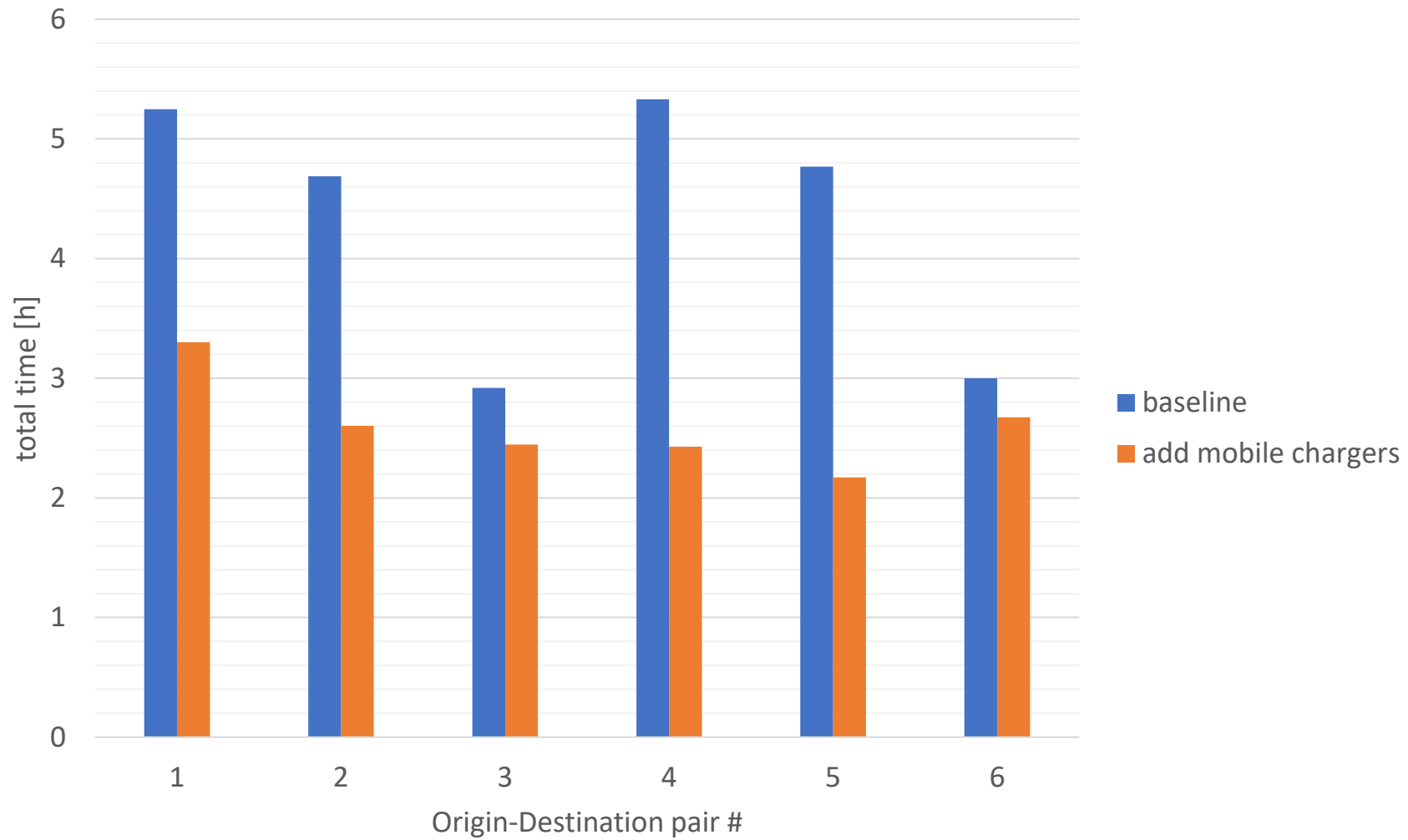
Baseline: Minimizing evacuation time **without** the aid of **mobile charging station**, and assume **constant vehicle driving speed**



Optimized: Minimizing evacuation time **with** the aid of **mobile charging station**, and assume the number of **vehicle on the road affects driving speed**

- Evacuation Origin/Destination
- Location where EV is charged
- Location with existing charging station
- Location with new mobile charging station

Approach #2: Results



Questions

We need data to serve our communities

- Use case maps
 - **Node location** & road **distances** & **speed limit** & travel **efficiency**
 - Charger information: **number of ports**, preferably the **charging rate**
 - Measure of **free space** for **mobile charger** deployments for each node
- MCS examples
 - **Energy? Outlets? Wattage?**
- Existing evacuation planning methods (as baseline)

Q&A?

Improving Preparedness of Communities for Evacuations using ZEVs

Definition of use cases for Merced/Mariposa evacuations

Presenter: Qijian Gan (UC Berkeley)

April 26th, 2024

Problem Statement

- Due to the rapid increase of ZEV ownership in recent years, rural areas like in Mariposa are facing a big challenge in the evacuation using ZEVs.
 - These rural areas normally lack charging facilities to support mass evacuations using ZEVs.
 - The transportation networks are vulnerable to support mass evacuations.
 - Very limited routes to nearby cities.
 - These routes normally consist of state highways that are 2-lane roadways. Therefore, any traffic incident happening on the roadway will significantly degrade the evacuation effectiveness and efficiency.
 - Changes to traffic control at arterial intersections are needed to support the large volume of vehicles traveling in the dominated direction.

Problem Statement (cont.)

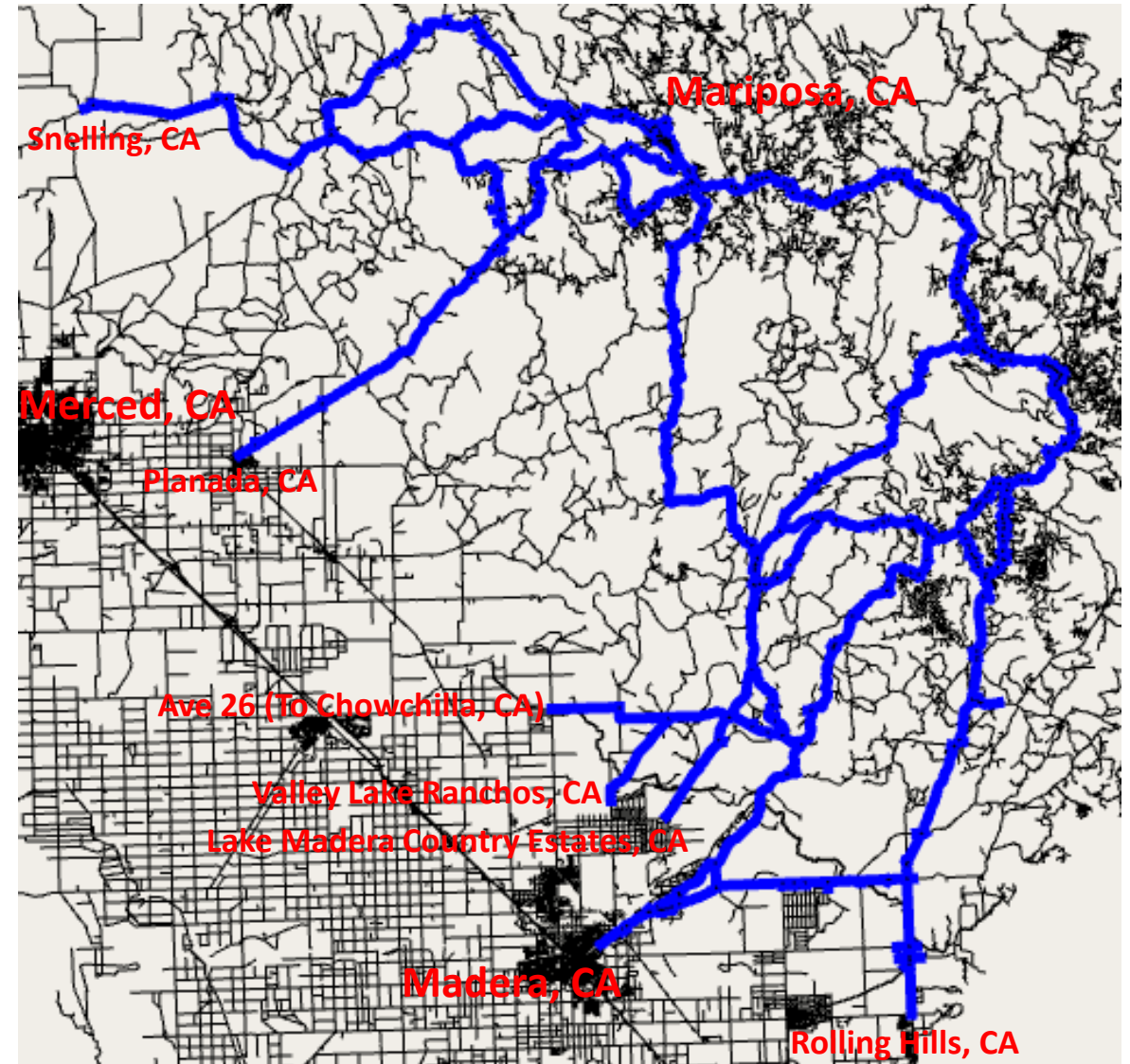
- Once the evacuation plans for ZEVs are developed (normally at the planning/macroscopic level), their performance needs to be evaluated at the operational/microscopic level.
 - The planning/macroscopic tools may not be able to appropriately capture
 - Abnormal traffic congestion at arterial intersections due to the factors like excessive left-turn demands and low intersection throughputs (e.g., at stop signs).
 - The impact of traffic incidents, e.g., reduced throughputs, longer travel delays, road blockages, gridlocks, etc.

Proposed Solution

- In this project, we want to develop a microsimulation platform in Aimsun (<https://www.aimsun.com/>) to support the evaluation of ZEV evacuation plans in the Merced/Mariposa area, which consists of the following:
 - A calibrated microsimulation model for the Merced/Mariposa network.
 - An enhanced simulation platform with novel features to support
 - The modeling and calculation of EV battery consumption in a more dynamic and realistic environment;
 - The modeling and management of fixed/mobile charging stations;
 - The routing of EVs for charging;
 - The modeling and assessment of evacuation plans (e.g., activation/deactivation of new traffic control along the evacuation routes, vehicle routing, performance measures, etc.)

Current Progress

- We have created a microsimulation network in the Merced/Mariposa area.
- While creating the network,
 - We focused on the scenario of ZEV evacuations from Mariposa to nearby cities (e.g., Merced, Madera, etc.)
 - We modeled the major arterials inside Mariposa, the state highways, and other 2-lane roadways.



Current Progress (cont.)

- We are in the process of calibrating this microsimulation model.
- We need your support if you happen to have or know how to get the following types of data for the study site:
 - Travel demand models (for Origin-Destination demands)
 - Timing sheets and traffic counts (or the Synchro model) at the signalized intersections
 - Traffic counts (or even the AADT) along the state highways and the 2-lane roadways (we might be able to get some data from the PeMS system)
 - Other data like probe travel times

Current Progress (cont.)

- Meanwhile, we have finished the architecture design for the modeling of EV charging at fixed/mobile charging stations in Aimsun.
- We are currently implementing the required data structures and workflows.
- We hope to show this enhanced simulation platform in our next workshop.

Q&A?